

Jaebin Jang, Ph.D.

Postdoctoral Research Associate

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Research profile

I study how environmental change shapes planktonic food webs in estuaries, coastal waters, and continental shelves. My research connects phytoplankton production and nutrient uptake with grazing, food sources, and trophic transfer.

I combine field observations, incubation experiments, stable isotopes, and statistical analysis to understand marine ecosystems, with a focus on the South Sea of Korea and the northern East China Sea.

Research appointment

2026-present · Postdoctoral Research Associate

Department of Environment and Energy Engineering
Gwangju Institute of Science and Technology (GIST)

Education

2019-2026 · Ph.D., Engineering

Environment and Energy Engineering
Gwangju Institute of Science and Technology (GIST)
Advisor: Prof. Chang-Keun Kang

2017-2019 · M.S., Earth Sciences and Environmental Engineering

Gwangju Institute of Science and Technology (GIST)

2013-2017 · B.S., Life Science

Gwangju Institute of Science and Technology (GIST)

Doctoral dissertation

Characterization of Processes Shaping Shallow Coastal and Shelf Planktonic Food Web Dynamics in the South Sea of Korea

Research projects

2024-present · Aquaculture carrying capacity for Pyropia farming

Manho Bay, Haenam

National Institute of Fisheries Science (NIFS)

Assessment of aquaculture carrying capacity for Pyropia farming in Manho Bay.

2022-present · Sustainable oyster production & carrying capacity

Northwestern Jinhae Bay

National Institute of Fisheries Science (NIFS)

Assessment of sustainable aquaculture productivity and carrying capacity for the Pacific oyster, *Crassostrea gigas*.

2022-2024 · Marine ecosystem responses to climate change

Tsushima Warm Current system

Korea Institute of Marine Science & Technology Promotion (KIMST)

Research on ecosystem shifts in Korean waters influenced by the Tsushima Warm Current.

2021 · Ocean warming, harmful phytoplankton & nutrient limitation

Experimental phytoplankton ecology

National Institute of Fisheries Science (NIFS)

Culture experiments, carbon and nitrogen uptake experiments, HPLC pigment analysis, and microscopy for research on nutrient limitation and biotoxin accumulation.

2017-2021 · Long-term marine ecosystem research

Korean waters influenced by the Tsushima Warm Current

Korea Institute of Marine Science & Technology Promotion (KIMST)

Field-based research on environmental change and marine ecosystem responses.

Training & honors

Croucher Summer Course

Climate Change and Marine Ecosystems, Hong Kong University of Science and Technology, Hong Kong

Summer session extension

University of California, Berkeley

Korean Government Sponsored Student

Gwangju Institute of Science and Technology, 2013-2025

Methods & experience

Field observations & incubations

Oceanographic surveys, seawater and plankton sampling, phytoplankton cultures, ^{13}C primary production, ^{15}N nitrate and ammonium uptake, dilution experiments, and mesozooplankton grazing.

Stable isotopes & fatty acids

EA-IRMS, biological and particulate sample preparation, fatty-acid extraction, FAME derivatization, and gas chromatography. Fatty-acid analyses contributed to Seo et al. (2023).

Plankton & biogeochemistry

Genus-level phytoplankton microscopy with Lugol fixation and Utermöhl sedimentation, ciliate and rotifer identification, HPLC pigments, CHEMTAX, chlorophyll a, nutrients, and particulate organic matter.

Reproducible analysis in R

GAMs (mgcv), regression, cross-validation, PCA, CCA, PERMANOVA, community and time-series analysis, Bayesian isotope mixing models (simmr), and isotopic niche analysis (SIBER).

Collaborative ecosystem modeling

Contributions to and interpretation of ROMS-based physical-biological and NPZD modeling for aquaculture carrying-capacity assessment. Additional software: PRIMER, SPSS, and SigmaPlot.

Mentoring & research practice

Regular experimental guidance and practical R analysis support for students and junior colleagues throughout doctoral and postdoctoral research.

Conference presentations

2025 • Oral

Linking riverine discharge to shifts in temperate estuarine planktonic food web dynamics.
PICES Annual Meeting, Yokohama, Japan.

2024 • Poster

Effects of riverine discharge on planktonic food web in a temperate estuarine bay.
5th Asian Marine Biology Symposium, Bangkok, Thailand.

Peer-reviewed publications

Jang J. is shown in bold. † Equal contribution as first authors.

Jang J., Chen M., Kang H.Y., Kim C., Jang H., Kang Y., Kang C.K. (2026). Linking riverine discharge to shifts in temperate estuarine planktonic food web dynamics. *Marine Environmental Research*, 213, 107629. doi:10.1016/j.marenvres.2025.107629

Kim C.†, **Jang J.**†, Yang J., Choi J.H., Kang H.Y., Moon S.Y., Soh H.Y., Kim H., Kang C.K. (2025). Trophic niche shifts in Japanese anchovy, *Engraulis japonica*, during ontogenetic migration in a temperate continental shelf system. *Progress in Oceanography*, 234, 103471. doi:10.1016/j.pocean.2025.103471

Kang H.Y., **Jang J.**, Lee B.G., Jang H., Park H.J., Kang C.K. (2026). Algal resource availability shapes trophic convergence of deposit feeders in a low-turbidity dredged estuary. *Marine Pollution Bulletin*, 228, 119552. doi:10.1016/j.marpolbul.2026.119552

Kundu G.K., Kim C., **Jang J.**, Lee C.I., Kim D., Lim W.A., Choi J.H., Kang C.K. (2024). Seasonal water-column structure drives the trophic niche of fish communities on a temperate continental shelf. *Biology*, 13(12), 1041. doi:10.3390/biology13121041

Kang H.Y., Lee B.G., Park S.R., Kim C., **Jang J.**, Kang C.K. (2024). Trophic niche overlap in coralline algae- and coral-dominated rocky-bottom subtidal communities. *Marine Pollution Bulletin*, 208, 116997. doi:10.1016/j.marpolbul.2024.116997

Park H.J., **Jang J.**, Kim D., Lee C.I., Kim C., Lim W.A., Choi J.H., Kang C.K. (2024). Trophic assessment of red seabream (*Pagrus major*) as a possible bioindicator of human activities in the South Sea of Korea using stable C and N isotopes. *Marine Pollution Bulletin*, 200, 116146. doi:10.1016/j.marpolbul.2024.116146

Kim D., Ji R., Feng Z., **Jang J.**, Lee D.I., Lee W.C., Kang C.K. (2023). Estuarine dam water discharge enhances summertime primary productivity near the southwestern Korean coast. *Marine Pollution Bulletin*, 191, 114971. doi:10.1016/j.marpolbul.2023.114971

Peer-reviewed publications (continued)

Seo D., Kim C., **Jang J.**, Kim D., Kang C.K. (2023). Identification of phytoplankton-based production of the clam *Corbicula japonica* in a low-turbidity temperate estuary using fatty acid and stable isotope analyses. *Water*, 15(9), 1670. doi:[10.3390/w15091670](https://doi.org/10.3390/w15091670)

Park T.H., **Jang J.**, Chun C.K., Lee Y., Han K.S., Yang S.J., Park H.J. (2023). Trophic assessment of an artificial kelp *Eisenia bicyclis* bed off the eastern coast of Korea based on stable isotope analyses. *Water*, 15(6), 1099. doi:[10.3390/w15061099](https://doi.org/10.3390/w15061099)

Kim D., Ji R., Park H.J., Feng Z., **Jang J.**, Lee C.I., Kang Y.H., Kang C.K. (2021). Impact of shifting subpolar front on phytoplankton dynamics in the western margin of East/Japan Sea. *Frontiers in Marine Science*, 8, 790703. doi:[10.3389/fmars.2021.790703](https://doi.org/10.3389/fmars.2021.790703)

Bibi R., Kang H.Y., Kim D., **Jang J.**, Kim C., Kundu G.K., Kang C.K. (2021). Biochemical composition of seston reflecting the physiological status and community composition of phytoplankton in a temperate coastal embayment of Korea. *Water*, 13(22), 3221. doi:[10.3390/w13223221](https://doi.org/10.3390/w13223221)

Kang H.Y., Kim C., Kim D., Lee Y.J., Park H.J., Kundu G.K., Kim Y.K., Bibi R., **Jang J.**, Lee K.H., Kim H.W., Yun S.G., Kim H., Kang C.K. (2020). Identifying patterns in the multitrophic community and food-web structure of a low-turbidity temperate estuarine bay. *Scientific Reports*, 10, 16637. doi:[10.1038/s41598-020-73628-6](https://doi.org/10.1038/s41598-020-73628-6)

Bibi R., Kang H.Y., Kim D., **Jang J.**, Kundu G.K., Kim Y.K., Kang C.K. (2020). Dominance of autochthonous phytoplankton-derived particulate organic matter in a low-turbidity temperate estuarine embayment, Gwangyang Bay, Korea. *Frontiers in Marine Science*, 7, 580260. doi:[10.3389/fmars.2020.580260](https://doi.org/10.3389/fmars.2020.580260)

Manuscripts in review & submitted

Status as of September 2026. These manuscripts are listed separately from peer-reviewed publications.

Jang J., Kang H.Y., Zang Z., Ji R., Kim D., Jang H., Kwak J.H., Kang C.K. Hydroclimatic forcing and oyster nutrient recycling jointly regulate new and regenerated production in a monsoon-influenced embayment. Marine Pollution Bulletin. **Under review.**

Kang H.Y., **Jang J.**, Kang Y., Park H.J., Yu O.H., Kang C.K. Phytoplankton size structure shifts benthic carbon pathways but conserves trophic architecture on continental shelves. Limnology and Oceanography Letters. **Submitted.**

Kang H.Y., **Jang J.**, Park S.E., Jang H., Lim W.A., Kang C.K. Metabolic stress under marine heatwaves mediates longline-culture oyster mortality. Aquaculture. **Submitted.**

Coauthored conference presentations

Kang H.Y., Lee Y.J., Yun S.G., Kim C., Park H.J., Kim Y.K., Kang Y., Kundu G.K., Kim D., **Jang J.**, Moon C.H., Kang C.K. (2019). Patterning Multitrophic Community and Trophic Structure in a Temperate, Low-turbidity Estuarine Bay of Korea. 16th Annual Meeting of Asia Oceania Geosciences Society, Singapore. Poster.

Bibi R., Kang Y., Lee Y.J., Kang H.Y., Kim D., **Jang J.**, Kang C.K. (2019). Seasonal variation of biogenic elements and phytoplankton biomass in a temperate bay of Korea subjected to low-turbidity river water discharge. 16th Annual Meeting of Asia Oceania Geosciences Society, Singapore. Poster.